

UNIT

5

Data Visualization

SYLLABUS

Data Visualization: Pixel-oriented Visualization Techniques, Geometric Projection Visualization Techniques, Icon-based Visualization Techniques, Hierarchical Visualization Techniques, Visualizing Complex Data and Relations.

LEARNING OBJECTIVES

- ✓ Definition, Advantages and Application Areas of Data Visualization
- ✓ Overview of Pixel-oriented Visualization Technique
- ✓ Various Icon-based Visualization Techniques
- ✓ Two Common Hierarchical Visualization Methods
- ✓ Visualizing Complex Data and Relations

INTRODUCTION

Data Visualization is a process of converting numeric data into some meaningful images which are easily interpreted by the humans. In other words, it can be defined as a graphical representation of information that provides the viewer with qualitative interpretation of information. Various data visualization techniques include, pixel-oriented technique, geometric projection technique, icon-based technique, hierarchical visualization technique and tag cloud visualization technique.

5.1 PIXEL-ORIENTED VISUALIZATION TECHNIQUES

Q1. Explain in detail about data visualization and also list its major advantages.

Answer:

Data Visualization

Data Visualization is a process of converting numeric data into some meaningful images which are easily interpreted by the humans. In other words, it can be defined as a graphical representation of information that provides the viewer with qualitative interpretation of information. It is a study of visual representation, wherein the information is abstracted in some schematic format. Earlier, the field of data visualization was related to information graphics and statistical graphics but now it has become an important area in the disciplines of research, teaching and development.

The main purpose of employing this technique is to reduce the difficulty in understanding the textual data which is generally collected from numerous sources like satellites, surveys or computer simulations. The difficulty in understanding the textual data is because of its huge quantity and complexity. The current visualization technique depends on computer graphics in order to represent the data in an innovative way such that complete information becomes obvious as well as easy to understand via images and animations. The common visualization techniques include, charts, graphs, plots, maps, 3D surface etc.

Basically, a human brain has the ability of processing the visual information consisting of different physical objects quickly when compared to textual information. The primary function of data visualization tool is to help the users in examining the complex data sets. This analysis is done by considering the physical characteristic properties such as transparency, curvature, speed, color, lighting effects with regard to the data.

Data visualization technique has the capability of creating multidimensional structure and models from unprocessed data. These structures and models are generally used in decision support technologies. It is very easy to implement the data mining techniques when the required data is present in warehouse or in a multidimensional server database. The technique of data visualization is very powerful because:

1. Human perception is dominated by human visual cortex.
2. Humans visually convert physical objects into information quickly than converting textual information into physical objects.

Advantages of Data Visualization

Following are the advantages of data visualization,

1. It helps in identifying the hidden patterns present within the data.
2. It helps in developing polling applications that enables the humans to focus only on the important issues.
3. It helps the organization to gain a competitive advantage by identifying the important trends in corporate and market data.

Q2. List the typical application areas of data visualization techniques.

(OR)

List out various applications of data visualization.

Jan./Feb.-23(R18), JNTU-H, Q10(b), M[5]

Answer:

Application areas of data visualization techniques are as follows,

1. Data visualization is used in the telecommunication areas where the visualized information helps in performing the following:
 - (i) Managing the network operations.
 - (ii) Managing the call-center environment.
 - (iii) Analyzing the service policy.
 - (iv) Analyzing the market status.
 - (v) Analyzing the loaded patterns.
2. It is used in the retail businesses where visualized information helps in performing the following:
 - (i) Analyzing the customer/product cross-selling.
 - (ii) Creating a strategy to determine the product's price.

3. It is used in banking areas where visualized information helps in performing the following:
 - (i) Managing the E-banking
 - (ii) Determining the risks obtained through credit card transactions.
4. It is used in insurance areas where visualized information helps in performing the following:
 - (i) Managing the assets and liabilities.
 - (ii) Modeling and calculating the insurance risks and premiums.
 - (iii) Managing and analyzing the workflows.
5. It is used in the government areas where visualized information helps in performing the following:
 - (i) Analyzing the budget.
 - (ii) Managing the resources.
 - (iii) Analyzing the economic cost.
 - (iv) Detecting the illegal actions.
6. It is used in the transportation areas where visualized information helps in performing the following:
 - (i) Analyzing the outputs.
 - (ii) Utilizing the assets.
 - (iii) Managing the movements.
7. It is used in capital market areas where visualized information helps in performing the following:
 - (i) Evaluating the risks.
 - (ii) Trading the derivatives.
 - (iii) Marketing the retail products.
 - (iv) Creating institutional sales systems.
8. It is used in the health care and medical areas where visualized information helps in performing the following:
 - (i) Analyzing the remedies/therapies.
 - (ii) Analyzing claims, i.e., the request made for compensation in terms of an insurance policy.
9. It is used in asset management areas where visualized information helps in performing the following:
 - (i) Analyzing portfolio performance.
 - (ii) Optimizing portfolios.
 - (iii) Allocating assets.

Q3. Discuss in brief about various participants in data visualization.

M[5]

Answer:

Data visualization is applied to mainstream computing and intelligent visualization. The main stream computing integrates the data visualization with decision support tools and applications. The intelligent visualization performs data interpretation. Along with decision support tools, most of the OLAP vendors provide three-dimensional visualization tools also. Consider for example Forest Tree 6.0 that is a web enabled development tool containing three-dimensional visualization version that allows the users to visualize as well as manage several dimensions in single view. Certain new visual tools are being developed in order to analyze web site performance. An example of it is ADVIZOR solutions.

Dashboards and Scorecards

The executives who are busy mostly require data visualization. Charts, tables and graphs are loaded into the EIS of 1990's. Later on they evolved into management cockpit products, dashboards and scorecards.

Visual Analysis

The nontechnical users who are gathering information from business data can perform enterprise data analysis. An example of it is visual, a database query language which empowers Hyperion's visual explorer. Most of the companies provide tools for visual analysis that can be performed interactively.

Financial Data Visualization

The BI data visualization is applied even in the area of finance. The Chief Financial Officers (CFO's) need an assurance that the processing power of computer is tempered with human being insight. This is required to

prevent the systems from identifying the patterns that are meaningless in data. Data visualization can be used for performing such task. For example a program is developed through which CFO can fly on three-dimensional landscape. This will represent risk, return and liquidity of company's assets. Laster-tree dimensional visualization is used in cyberspace.

Q4. Discuss in brief about pixel-oriented visualization technique.

M[5]

Answer:

Pixel Oriented Visualization Technique

In the technique of pixel oriented visualization the dimensional value is visualized using a colored fixed. Here, the pixel's color represents the value of dimension. In this technique, if data set contains 'k' dimensions then exactly "k" windows are built on the screen. This technique maps k dimensions value to 'k' colored pixels which one placed at respective positions in the window.

The concept of pixel-oriented visualization technique can be better understood by the following example. Consider 'Bajaj Electronics' store. It maintains the information of customer in 'customer-info' table where the table contains five dimensions namely remuneration, credit-card-amount, No-of-transactions, Age and Cust-Id. The relationship between remuneration and remaining attributes can be analyzed using the concept of visualization. Initially sort the remuneration of all customers in descending order. Then, arrange the information pertaining to customers in five visualization windows by following the remuneration order for all customers. The color of the pixel is selected in such a way that higher values are represented with darker shade and smaller values are represented with lighter shade.

The following figure illustrates pixel-oriented visualization for five dimensions.

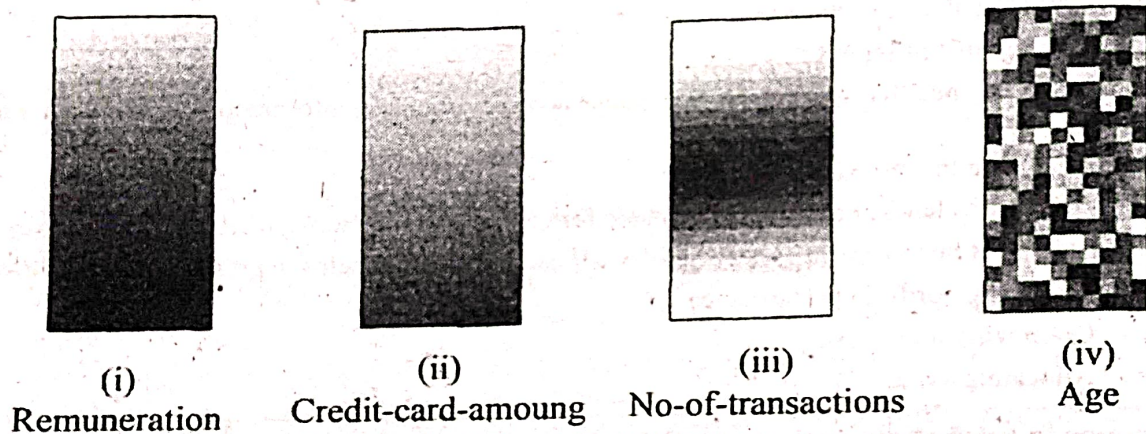


Figure: Pixel-oriented Visualization for Five Attributes

From the above figures (i) and (ii) can be inferred that, CreditCardAmount is directly proportional to remuneration. In worlds as credit card amount increases the remuneration also increases. In figure (iii) it is observed that customers who purchase more from the store are having the remuneration that fall in the middle range and finally figures (iv) and (v) conveys that there is no relationship between remuneration and cust ID and remuneration and age, respectively.

5.2 GEOMETRIC PROJECTION VISUALIZATION TECHNIQUES

Q5. Explain geometric projection visualization techniques.

March-24(R18), JNTU-H, Q10, M[10]

(OR)

Demonstrate Geometric projection visualization techniques. Jan./Feb.-23(R18), JNTU-H, Q10(a), M[5]

Answer:

In pixel-oriented visualization technique, the distribution of data in multidimensional space cannot be specified clearly. This drawback is overcome in geometric projection visualization. The objective of geometric projection visualization technique is to locate or find geometric transformations and projections of multidimensional data sets. It mainly focuses on visualizing multidimensional space on two-dimensional (2-D) display.

The various techniques of geometric projection visualization includes the following.

- (i) Scatter plot
- (ii) Scatter plot matrix
- (iii) Parallel co-ordinates

In scatter plot technique, 2-D data points are represented using a pair of co-ordinates. It includes third dimension by using different colors or shapes.

The following figure illustrates visualization of two-dimensional (2-D) data set using scatter plot technique.

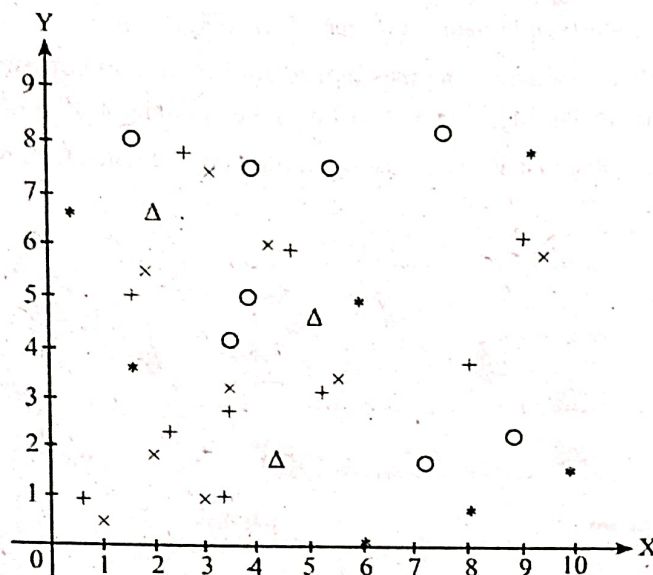
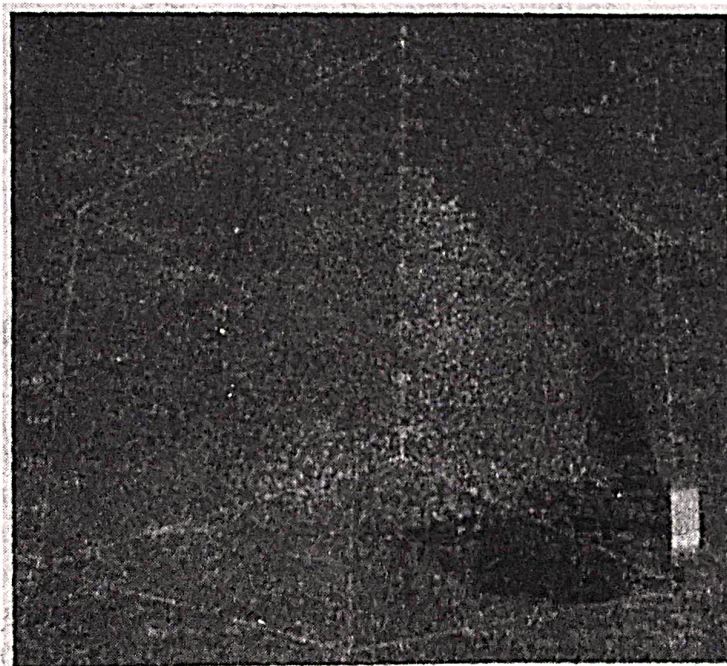


Figure: 2-D Data Set Visualization Using a Scatter Plot

In the above figure, 'X' and 'Y' represents spatial attributes and third dimension is displayed using different shapes.

In three dimensional (3-D) scatter plot technique three axes viz X, Y and Z are used in cartesian co-ordinate system for displaying data points like scatter plot technique, 3-D scatter plot can also add additional by using different colors. Thus, 3D scatter plot can display upto four dimensional data points.

The flowing figure illustrates visualization of 3-D data set using a scatter plot technique.



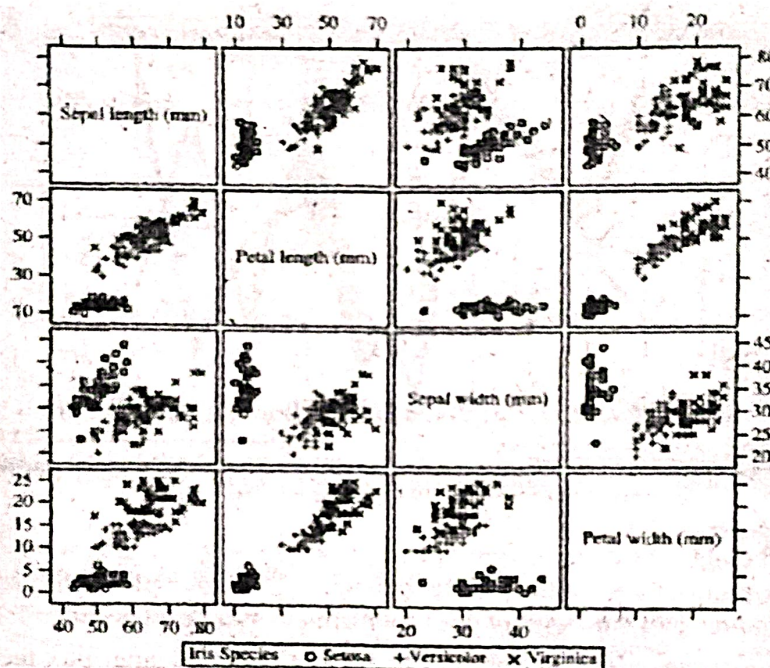
Figure

The technique of scatter plot are inefficient if the number of dimensions in a data set are greater than four. So, an enhanced technique of scatter plot called scatter-plot matrix is used. In this technique, if data set contains 'k' dimensions then $k \times k$ grid of 2-D scatter plots represents a scatter-plot matrix. This provides a visualization for each dimension with remaining dimensions.

Like scatter plot technique, scatter-plot matrix method also becomes inefficient with the increase in dimensionality. So, in order to deal with higher dimensionality, a technique called 'parallel co-ordination' is used. In this technique, a data set containing 'k' dimensions are visualized by drawing 'n' equally spaced axes. Where each axis represents one dimension and it is parallel to one of the display axes.

The technique of parallel co-ordinates becomes inefficient for data set containing several records as visual cluster and overlap decreases the readability of visualization. Thus, making it difficult to locate the patterns.

The following figure illustrates visualization using parallel co-ordinates technique.



Figure

5.3 ICON-BASED VISUALIZATION TECHNIQUES

Q6. Discuss the two popular icon-based visualization techniques.

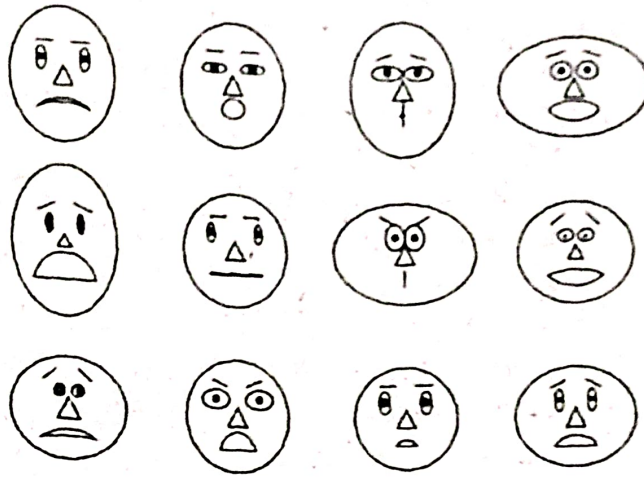
M[5]

Answer:

Icon-based Visualization Techniques

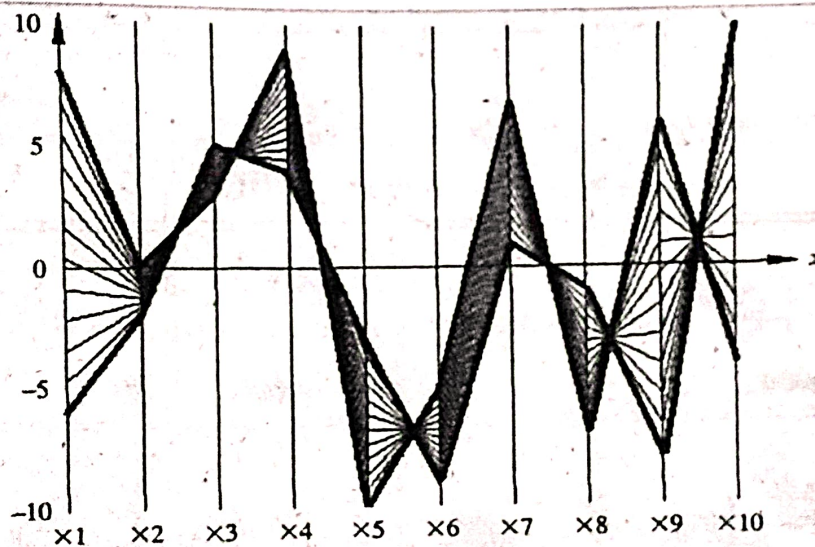
Icon-based visualization techniques makes use of small icons for representing multidimensional data values. Chernoff faces and stick figures are the two popular icon based techniques.

- (i) **Chernoff Faces:** This technique was implemented by statistician Herman Chernoff in the year 1973. It displays multidimensional data in the form of cartoon human face upto 18 dimensions. It specifies the dimensional value of various components like eyes, ears, mouth and nose by their size, shape, position and orientation. Moreover, it also utilizes the mindset of a human in identifying the differences between facial features. However, large tables of data cannot be viewed easily using this techniques. This is because, the facial expressions can differ based on the order in which dimensions are allotted to facial characteristics. In addition to this, some of the data values may not be shown due to limited dimensional value. In order to overcome this issue, the enhanced version of Chernoff faces technique known as 'Asymmetrical Chernoff Faces' came into existence. This technique displays multidimensional data upto 36 dimensions.



Figure

- (ii) **Stick Figure:** This technique maps multidimensional data to five-piece stick figures in such a way that every figure contains four limbs and one body. Here, two dimensions are mapped to x and y-axis and the rest of them are mapped to length of the limbs.



Figure

5.4 HIERARCHICAL VISUALIZATION TECHNIQUES

Q7. Explain various hierarchical visualization techniques.

M[10]

(OR)

How to perform visualization of the data using a hierarchical partitioning into subspaces?

Explain with examples.

Feb.-22(R18), JNTU-H, Q8, M[15]

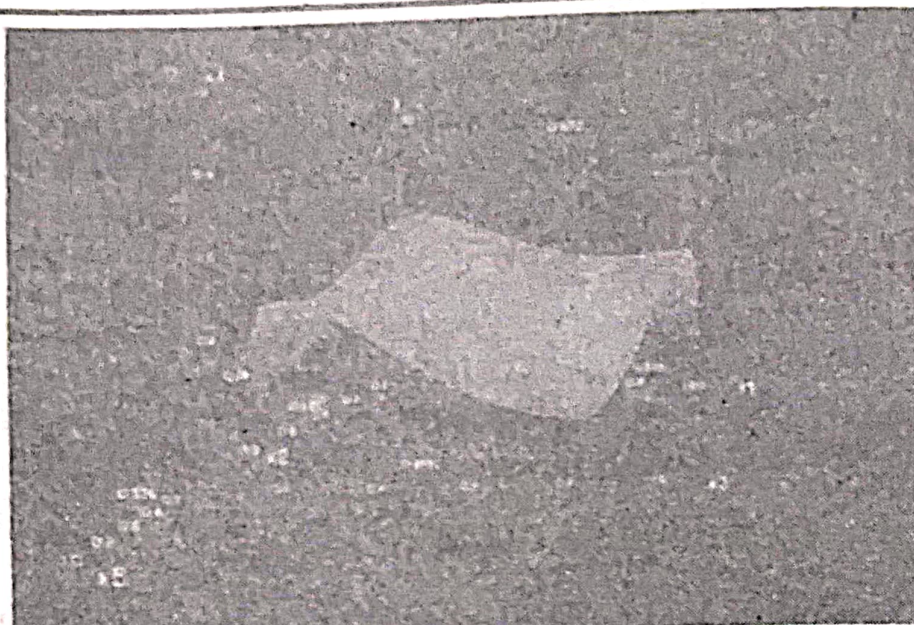
Answer:

In this type of visualization, all dimensions are further divided into subsets in hierarchical form. Worlds-within-worlds and tree maps are the two hierarchical visualization methods.

(i) **World's-Within-World's**

It is a representative hierarchical visualization method. Consider a 6D-data set with dimensions $A, x_1, x_2, \dots, x_5$. In order to examine the impact of dimensions x_1, x_2, x_3, x_4 and x_5 on dimension A let us assign some specific value say b_3, b_4, b_5 to x_3, x_4, x_5 . Now, visualize A, x_1, x_2 , by means of a 3D-plot as follows,

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Figure

Place the position of the inner world origin at points b_3, b_4, b_5 in the outer world. Now, the user can vary the dimensions and position of the inner world origin in the outer world. Similarly, user can use multiple worlds and so it is referred as “world-within-worlds”.

- (ii) **Tree-Maps:** This technique displays hierarchical data in the form of group of nested rectangles. The following figure illustrates a treemap visualizing google news stories,



In the above figure, all news stories are arranged into seven types. Each type of stories further divided into subcategories thus forming a tree-map.

Q8. Discuss about dimensional stacking and tree-map.

March-21(R18), JNTU-H, Q6(b), M[7]

Answer:

Dimensional Stacking

Dimensional stacking refers to a specialized technique used to display multi-variate data in two-dimensional form. To do this, it splits the axes and generates multiple divisions. The count of divisions generated can be identified using cardinality of dimension.

Consider the following dimensional information.

	Minimum	Maximum	Cardinality
Customers	0	20	4
Managers	0	30	4
Cashiers	0	20	4
Salesmans	0	20	4

The above table can be represented in the form of nested grids as follows.

The rectangles in the grid represents values where the highlighted one represents ranges. Filling is done only if there exists certain record associated with ranges of all dimensions.

Dimensional stacking user limited rendering area and therefore, it is highly unscalable.

Tree-map

For answer refer Unit-V, Q7, Topic: Tree-Maps.

5.5 VISUALIZATION COMPLEX DATA AND RELATIONS

Q9. Describe the tag cloud visualization technique for complex data and relations.

M[5]

(OR)

Explain the challenges in visualizing complex data and relations and suggest suitable mechanisms to address them.

March-24(R18), JNTU-H, Q11, M[10]

(OR)

Explain about tag cloud visualization technique.

Jan./Feb.-23(R18), JNTU-H, Q11(b), M[5]

(OR)

Discuss in detail about visualizing complex data and relations: March-21(R18), JNTU-H, Q7(b), M[7]

(OR)

Explain the challenges in visualizing complex data and relations and suggest suitable mechanisms to address them.

Sep.-20(R16), JNTU-H, Q8, M[15]

Answer:

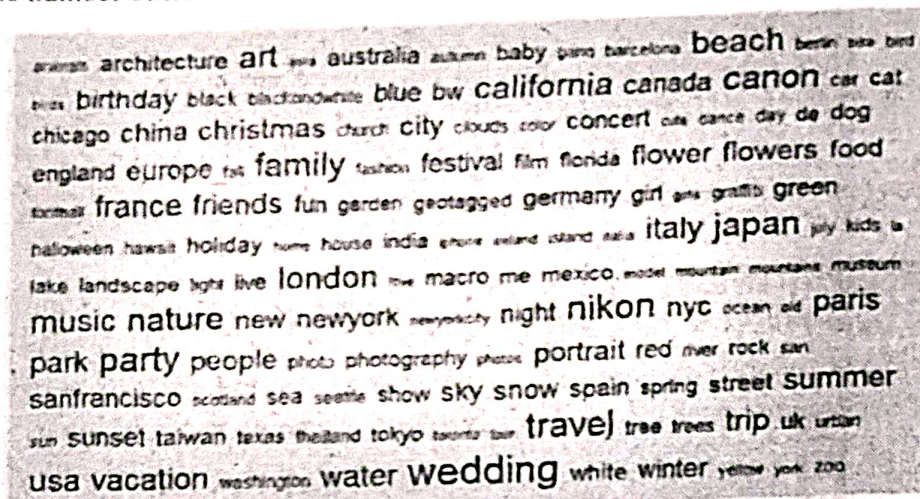
Visualizing Complex Data and Relations

Earlier, visualization techniques were generally used for numeric data. But, at present most of the new visualization techniques can be applied on complex data and relations. A tag cloud is one such technique.

A tag cloud is a visualization technique which is used for tagging various objects like images, blog entries and product reviews on web. In this technique, tags are listed in either alphabetical order or in a user preferred order. The essentiality of a tag is specified by specific font size or font color. The following figure illustrates tag cloud for visualizing popular tags used in a web site.

A tag cloud can be utilized in either of the ways.

- (i) To define the number of times a tag is applied on a item by distinct users.
- (ii) To define the number of items on which a tag is applied.



Q10. Write down steps involved in Data Visualization in Tableau. Jan./Feb.-23(R18), JNTU-H, Q11(a), M[5]

Answer:

Data Visualization in Tableau

The procedure for data visualization in Tableau involves the following steps,

1. **Data Extraction:** Data extraction requires selection of measures that are numerical entities (like visits, bounces) and dimensions that are categorical type data points (like source medium, landing page). The number of dimensions should be kept minimum as large number of dimensions results in large data sets.
2. **Workspace Knowledge:** The basic workspace of the tool carries 3 sections, namely data (containing dimensions and measures), settings (containing pages, filters and marks) and visualization. Also, the top of the screen carries rows and column sections.

For data visualization, the user has to drag and drop the icons from dimensions and measures into the columns and rows area.

Example

If the user wants to check the traffic data by 'medium' dimension, then he/she needs to drag the medium icon from dimensions to columns and visits icon from measures to rows. The user can drag the date icon to the end of column line for obtaining historic performance. For better understanding of visualization, the user can change the date menu to months.

3. **Data Enhancements:** Tableau offers enhanced data by offering a special functionality called "calculated field". Using this, users need not have to bother about the differences existing among dimensions and measures.

If the user wants to calculate the bounce rate, the formula (Bounces/visits) can be entered in the selected 'calculated field' by right clicking the measures.

4. **Data Segmentation:** Tableau offers data segmentation by providing necessary parameters that can be used to divide the data accordingly. The required formulae are entered in the formula block to check the functionalities.

Example

Blog and non-blog pages bounce rate, source calculations.

5. **Data Filtration:** Tableau offers two methods for filtering the data. One method is to drag and drop the element (that needs to be filtered) into the filter area and selecting the filter option. Another method is to selecting the quick filter option after right clicking on any of the elements (that are to be filtered) and the filters are added to the right hand column.
6. **Quick Visualizations:** Tableau provides a feature 'show me' on the top right of the screen for quick visualizations. It displays various visuals of tableau and guides the user for using different options.

VERY SHORT QUESTIONS WITH SOLUTIONS (VSQS)**Q11. What is data visualization?****Answer:**

March-24(R18), JNTU-H, Q1(I), M[2]

Data Visualization is a process of converting numeric data into some meaningful images which are easily interpreted by the humans. Earlier, the field of data visualization was related to information graphics and statistical graphics but now it has become an important area in the disciplines of research, teaching and development.

Q12. What are the advantages of Data Visualization?**Answer:**

Jan./Feb.-23(R18), JNTU-H, Q1(J), M[3]

Following are the advantages of data visualization,

1. It helps in identifying the hidden patterns present within the data.
2. It helps in developing polling applications that enables the humans to focus only on the important issues.
3. It helps the organization to gain a competitive advantage by identifying the important trends in corporate and market data.

Q13. What is the application of data visualization in finance?**Answer:**

M[1]

The BI data visualization is applied even in the area of finance. The Chief Financial Officers (CFO's) need an assurance that the processing power of computer is tempered with human being insight. This is required to prevent the systems from identifying the patterns that are meaningless in data. Data visualization can be used for performing such task.

Q14. What is pixel-oriented visualization technique?**Answer :**

M[1]

In the technique of pixel oriented visualization the dimensional value is visualized using a colored fixed. Here, the pixel's color represents the value of dimension. In this technique, if data set contains 'k' dimensions then exactly "k" windows are built on the screen.

Q15. When scatter plot technique becomes inefficient? How to recover from this situation? M[1]**Answer:**

The technique of scatter plot are inefficient if the number of dimensions in a data set are greater than four. So, an enhanced technique of scatter plot called scatter-plot matrix is used. In this technique, if data set contains 'k' dimensions then $k \times k$ grid of 2-D scatter plots represents a scatter-plot matrix. This provides a visualization for each dimension with remaining dimensions.

Q16. List the ways in which tag cloud can be utilized.

M[1]

Answer:

A tag cloud can be utilized in either of the ways.

- (i) To define the number of times a tag is applied on a item by distinct users.
- (ii) To define the number of items on which a tag is applied.

Q17. What is Chernoff Faces technique?

Jan./Feb.-23(R18), JNTU-H, Q1(I), M[2]

Answer:

Chernoff faces technique was implemented by statistician Herman Chernoff in the year 1973. It displays multidimensional data in the form of cartoon human face upto 18 dimensions. It specifies the dimensional value of various components like eyes, ears, mouth and nose by their size, shape, position and orientation.

Q18. What are the benefits of using stick figures?

March-24(R18), JNTU-H, Q1(J), M[3]

Answer :

Stick figures technique maps multidimensional data to five-piece stick figures in such a way that every figure contains four limbs and one body. They help in making complex data easier to understand and analyze. Moreover, they can be used to retrieve motion data.

FREQUENTLY ASKED QUESTIONS & IMPORTANT QUESTIONS

Q1. Explain in detail about data visualization and also list its major advantages.

Important Question

Answer :

For answer refer Unit-I, Q1.

Q2. List the typical application areas of data visualization techniques.

Answer :

(Important Question | Jan./Feb.-23(R18), Q10(b))

For answer refer Unit-I, Q2.

Q3. Discuss in brief about pixel-oriented visualization technique.

Answer :

Important Question

For answer refer Unit-I, Q4.

Q4. Explain geometric projection visualization techniques. Describe geometric projection visualization technique.

Answer :

(March-24(R18), Q10) | Jan./Feb.-23(R18), Q10(a))

For answer refer Unit-I, Q5.



Q5. Discuss the two popular icon-based visualization techniques.

Answer :

Important Question

For answer refer Unit-I, Q6.

Q6. Explain various hierarchical visualization techniques.

Answer :

(Important Question | Feb.-22(R18), Q8)

For answer refer Unit-I, Q7.

Q7. Discuss about dimensional stacking and tree-map.

Answer :

(Important Question | March-21(R18), Q4(b))

For answer refer Unit-I, Q8.

Q8. Explain the challenges in visualizing complex data and relations and suggest suitable mechanisms to address them.

Answer :

(March -24(R18), Q11 | Jan./Feb.-23(R18), Q11(b) | March-21(R18), Q7(b) | Sep.-20(R16), Q8)

For answer refer Unit-I, Q9.



Q9. Write down steps involved in Data Visualization in Tableau.

Answer :

(Important Question | Jan./Feb.-23(R18), Q11(a))

For answer refer Unit-I, Q10.